Deep learning for biologists

The course, the instructors, the participants

Filippo Biscarini
Senior Scientist
CNR, Milan (Italy)

Nelson Nazzicari
Senior Scientist
CREA, Lodi (Italy)



Welcome to Deep Learning for Biologists!

- 5 days (mornings), from 9:30 to 12:30 + coffee break
- mix of:
    - slides → lectures & pills
    - jupyter notebooks → hands-on sessions

- introductory course aimed for:
    - biologists with little data analysis experience who wish to understand what all the fuss about DL is
    - more experienced biologists who wish to apply DL to their own research projects
    - a combination of the above



Welcome to Deep Learning for Biologists!

- the two instructors will interchange during the 5 days in leading the lectures and practicals: the other will assist you if you have questions or technical problems
- **questions are welcome at all times**: don't be shy and ask if something is not clear!
- the **hands-on sessions** will be a chance to practice what we learnt and discuss it together



Welcome to Deep Learning for Biologists!

- What is Deep Learning?
- What type of problems we will tackle
- The building blocks of deep learning
- From logistic/softmax regression to neural networks
- Cross-validation and performance measures
- More building blocks: advanced stuff
- Convolutional neural networks
- Transfer learning
- Embeddings, self-attention and transformers
- A bit of LLMs



Disclaimer

We haven’t used AI for any slides or hands-on session notebooks ;-)



Filippo in a slide

- **Roma** (*born*)
- **Perugia** (*MSc degree*)
- **Cork, ICBF** (*Web-design & Database*)
- **Cremona, ANAFI** (*Quantitative Genetics*)
- **Guelph, CGIL** (*Visiting Scientist*)
- **Wageningen, WUR** (*PhD*)
- **Göttingen University** (*post-doctoral researcher*)
- **Lodi, PTP** (*'omics in animals, plants, humans*)
- **Milan - CNR** (*tenured researcher*)
- **Cardiff University** (*biostatistician*)
- **Milan - CNR** (*senior researcher*)
- **Bruxelles - ERC** (*seconded national expert*)
- **Milan - CNR** (*director of research*)



Nelson in a slide

- **Tortona** (*born*)
- **Pavia** (*MSc computer engineering, PhD*)
- **Fairfax, GMU** (*informatics security*)
- **Lodi, PTP** (*database, biostatistician*)
- **Zagreb, Centre of Excellence for Biodiversity and Molecular Plant Breeding** (*biostatistician*)
- **Wageningen, NPEC@WUR** (*data scientist, drones*)
- **Lodi, CREA** (*senior researcher*)





Us on the internet

- https://bioinformateachers.github.io/
- https://github.com/ne1s0n/bioinformateachers



The participants

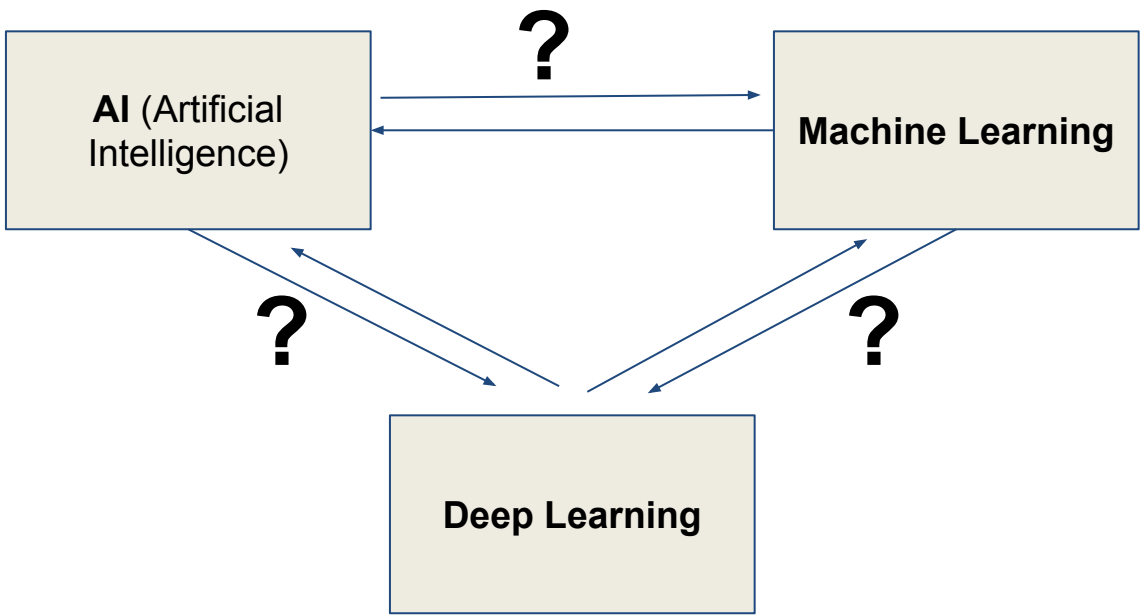
So, what about you?



A practical and theoretical introduction



What is deep learning?

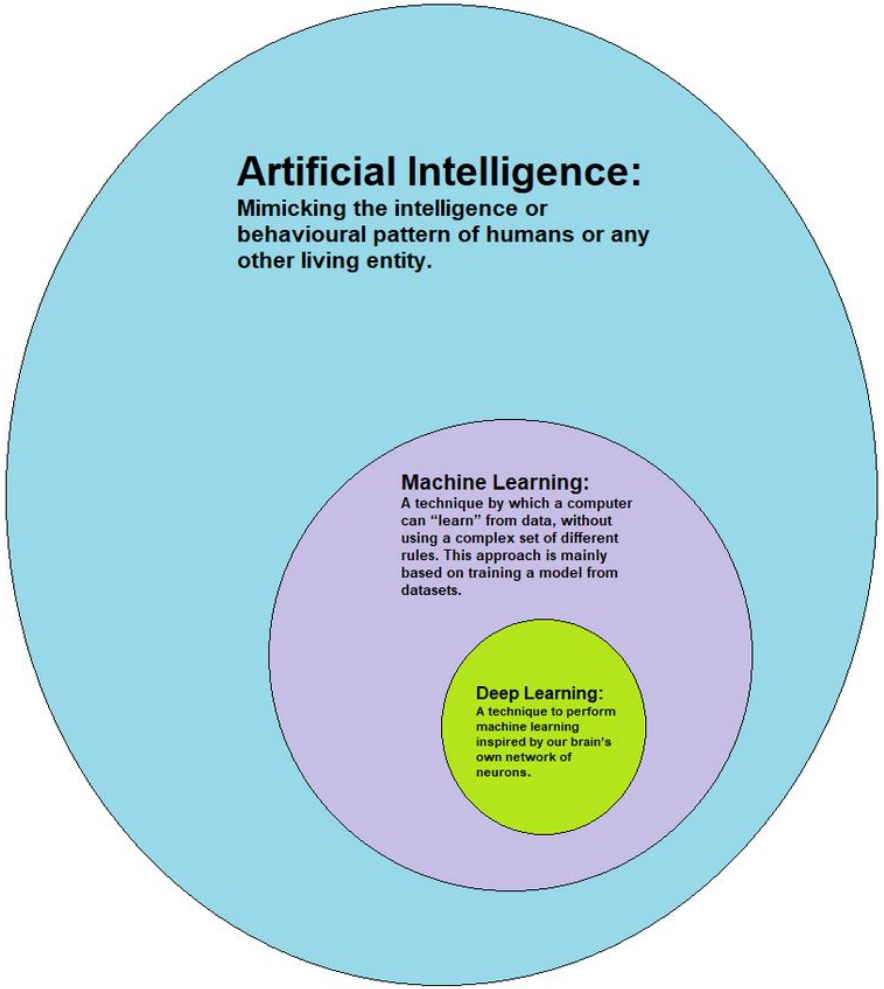




What is deep learning?

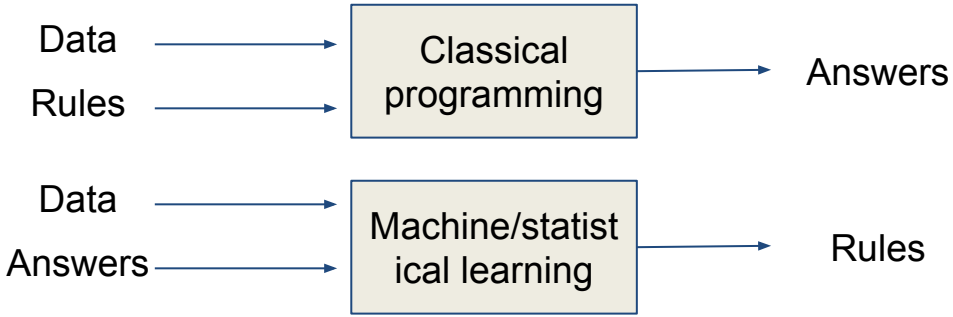
AI >> ML >> DL

Source: https://en.wikipedia.org/wiki/Deep_learning#/media/File:AI-ML-DL.png





What is (deep) learning?



a learning model is **trained** rather than explicitly programmed:

1. Input data (e.g. sound recordings, images)
2. Output examples (e.g. sound transcripts, image-tags)
3. Performance measure: how well is the algorithm working → adjustment steps → **learning**



The lucky gambler

You can do (statistical) learning in your head!

- You start with 1 euro
- After the first bet, you have 2 euros
- After the second bet, you have 4 euros
- After the third bet, you have 2 euros
- After the fourth bet, you have 1 euro
- **How many euros do you have after the sixth bet?**



The lucky gambler

You can do (statistical) learning in your head!

- You start with 1 euro
- After the first bet, you have 2 euros
- After the second bet, you have 4 euros
- After the third bet, you have 2 euros
- After the fourth bet, you have 1 euro
- **How many euros do you have after the sixth bet?**

you found a pattern! **(if the answer was correct)**



The lucky gambler

You can do (statistical) learning in your head!

TRAINING DATA

- You start with 1 euro (t = 0)
- After the first bet, you have 2 euros (t = 1)
- After the second bet, you have 4 euros (t = 2)
- After the third bet, you have 2 euros (t = 3)
- After the fourth bet, you have 1 euro (t = 4)
- **How many euros do you have after the sixth bet?**

NEW, UNKNOWN DATA

EUROS AFTER THE SIXTH BET = 4

PREDICTION

EUR = 1 x 2 exp(1 - cos(π*(t/2)))

MATHEMATICAL MODEL



What is deep learning?

- (machine; statistical) learning
  - tune a **mathematical model** using some **training data** to **make predictions** on unknown, new data
  - «If you make a bunch of random changes to your program until it sort-of works, that's "hacky" and "bad coding practice". But if you do it really fast, it's "machine learning"» [Steve Maine, Software Engineer]
- Deep learning
  - **Neural networks**, a mathematical model “inspired” by biology
  - Artificial Neural Networks have been around for ~80 years ...
  - … but became “deep” in the last ten years



Why “deep”?

What do you think?

(humorous) hint: think of “Deep Thought”’s answer = 42 :-)



Why “deep”?

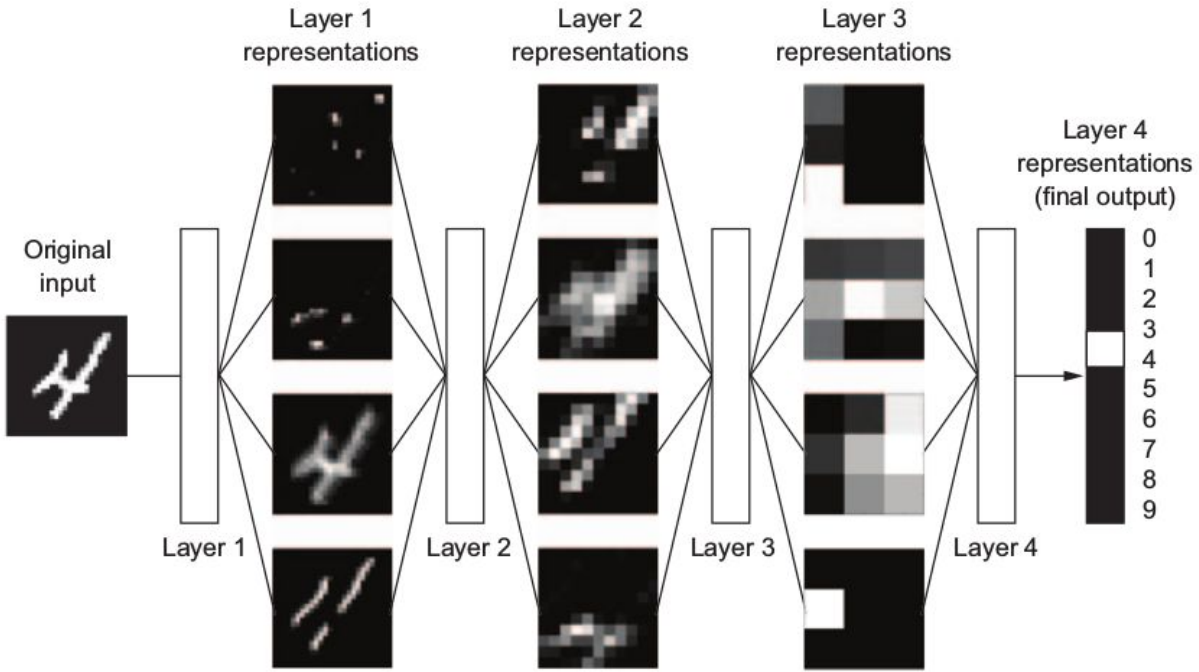


Figure 1.6 Deep representations learned by a digit-classification model

deep neural networks
(i.e. with many layers)



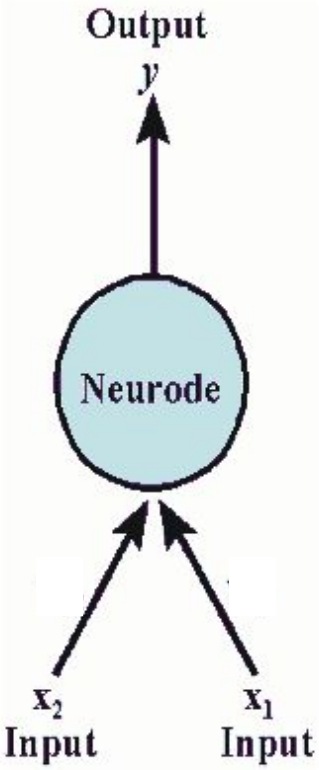
Source: François Chollet and J.J. Allaire “Deep learning with R” (2018)

A brief history of DL



How it all began

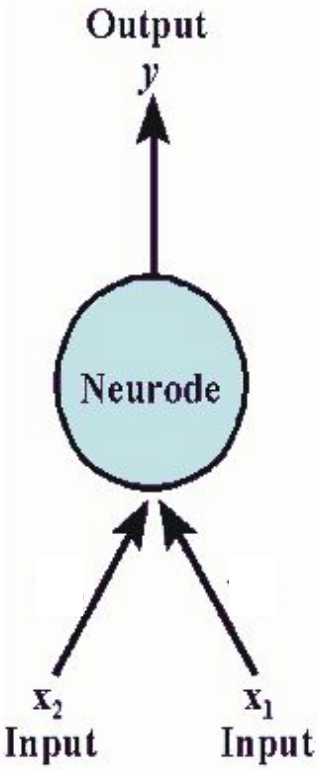
- **1943, McCulloch & Pitts**:
  - first mathematical model of a neural network
  - only binary input and output (0/1),
  - only used the threshold step activation function
  - did not incorporate weighting the different inputs
  - did not come out of the blue: they were trying to model the human brains (with the knowledge of the 1940's)
- **1950s**: pioneers started asking whether computers could be made to "think" → (A.I.)





How it all began

- **1943, McCulloch & Pitts**:
  - first mathematical model of a neural network
  - only binary input and output (0/1),
  - only used the threshold step activation function
  - did not incorporate weighting the different inputs
  - did not come out of the blue: they were trying to model the human brains (with the knowledge of the 1940's)
- **1950s**: pioneers started asking whether computers could be made to "think" → (A.I.)
- **1957, Rosenblatt**: introducing the **perceptron**:
  - weighting inputs
  - additional activation functions (e.g. sigmoid)





From symbolic AI to ML

- this was all **no learning AI**
  - e.g. early chess programs based on hardcoded rules crafted by programmers
  - experts believed that human-level A.I. could be achieved by handcrafting a sufficiently large set of explicit rules → **symbolic AI**
- dominant paradigm from the 1950s to the late 1980s
- **expert systems boom** of the 1980s
- symbolic AI suitable to solve well-defined, logical problems (like chess); impossible to figure out explicit rules for solving more complex, fuzzy problems like image classification, speech recognition etc.
- **enter learning**: ML emerged to take the place of symbolic AI
  - the missing piece was an efficient way to train large neural networks
  - → enter the **backpropagation algorithm**!
  - **1989, Yann LeCun**: first application of NN to the recognition of handwritten digits





Beware of the hype!

too high expectations (hype) → disappointment → < research investment (stop progress)

This happened twice in the past:

1. **symbolic AI in the 1960s**:
   a. expectations: 1967, Marvin **Minsky**: “the problem of creating ‘artificial intelligence’ will be solved soon”; Minsky, 1970: “in three to eight years we will have a machine with the general intelligence of an average human being”;
   b. failure: researchers and funds turned away from the field → **first AI winter**
2. **1980s**, **expert systems**:
   a. initial success stories attracted investment, companies developed expert systems
   b. expert systems were expensive to maintain, difficult to scale, limited in scope: interest declined → **second AI winter**



Beware of the hype!

too high expectations (hype) → disappointment → < research investment (stop progress)

This happened twice in the past:

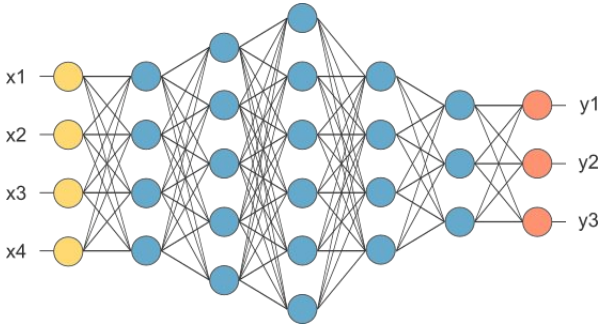
1. **symbolic AI in the 1960s**:
   a. expectations: 1967, Marvin **Minsky**: “the problem of creating ‘artificial intelligence’ will be solved soon”; Minsky, 1970: “in three to eight years we will have a machine with the general intelligence of an average human being”;
   b. failure: researchers and funds turned away from the field → **first AI winter**
2. **1980s**, **expert systems**:
   a. initial success stories attracted investment, companies developed expert systems
   b. expert systems were expensive to maintain, difficult to scale, limited in scope: interest declined → **second AI winter**

What now? Third cycle of AI hype and disappointment? In spite of possibly unrealistic short-term expectations, the long-term picture is bright: **no short-term hype → long-term vision instead!**



Rationing NN through winter, before summer!

- 1990s: **SVM** (Vladimir Vapnik, 1995)
- 2000s: **Random Forest, Gradient Boosting** (tree-based methods)
- **deep NN** (DL at last!):
  - 2011, Dan Ciresan: **GPU-trained deep neural networks** for image classification
  - 2012, Alex Krizhevsky & Geoffrey Hinton: **deep convolutional neural network** → 83.6% accuracy at image classification
  - DL automates feature extraction and engineering



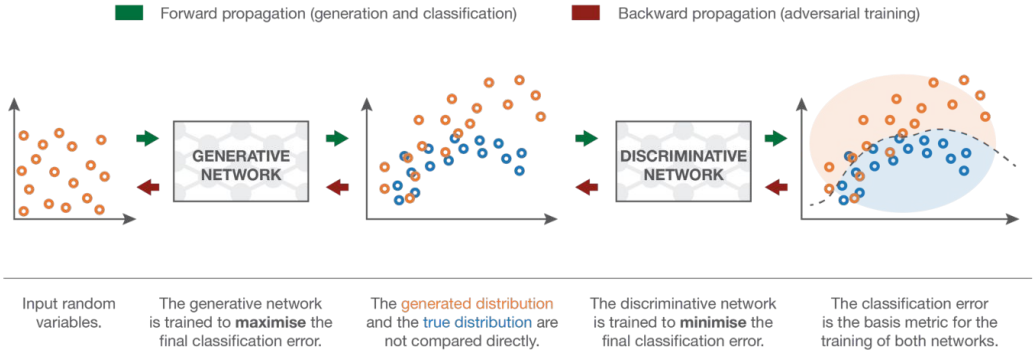
[From: https://github.com/ashishpatel26/Andrew-NG-Notes/blob/master/andrewng-p-1-neural-network-deep-learning.md]



Latest developments - what in the 2010/20s?

- DL and gradient boosting **dominate Kaggle competitions** since 2012
- **unsupervised / semi-supervised learning** → *Google Brain*, the cat experiment (2012, Arxiv)
- **Generative Adversarial Networks** (GANs): Ian Goodfellow et al., 2014: generative models trained "indirectly" by trying to fool another network that is trained at the same time to distinguish "generated" data from "true" data
- **GRU** (gated-recurrent units): Cho et al., 2014: breakthrough for **RNN**

- **LLMs**: large language models (more on this later)
- Self-attention ("Attention is all you need") →transformers
- GNN
- Agentic AI



[From: towardsdatascience.com]



[REF] History of deep learning

- McCulloch and Pitts, *A logical calculus of the ideas immanent in nervous activity*, 1943. https://link.springer.com/article/10.1007/BF02478259
- Wang and Raj, *On the Origin of Deep Learning*, 2017 https://arxiv.org/pdf/1702.07800.pdf
- Andrew Beam, Deep Learning 101 - Part 1: History and Background, 2017 https://beamandrew.github.io/deeplearning/2017/02/23/deep_learning_101_part1.html
- Chung et al., 2014. Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling → GRU layers https://arxiv.org/abs/1412.3555
- Hochreiter & Schmidhuber 1997. Long short-term memory → LSTM layers https://ieeexplore.ieee.org/abstract/document/6795963



Why now?

Innovations in:

- Hardware
- Big Data
- Algorithms
- Infrastructure



Why now?

Innovations in:

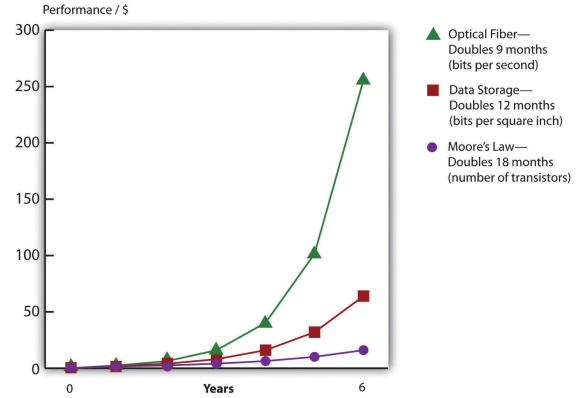
- Hardware:
  - CPUs now 5,000 times faster than 25 years ago
  - GPUs (thanks to videogames!)
  - TPUs (Tensor Processing Units: designed specifically for deep learning)
  - Active R&D on specialised A.I. processors/accelerators: e.g. Google's TPU, Intel's Loihi, IBM's TrueNorth, NVIDIA T4 GPU …



Why now?

Innovations in:

- Big Data:
  - Internet: search engines, social media, image datasets etc.
  - IoT: sensors, interconnected devices
  - Data storage (Moore's law)
  - GPT-3 was trained on 570 GB of text data (september 2021)
  - Google Search index: 1e+8 GB



https://2012books.lardbucket.org/books/getting-the-most-out-of-information-systems-v1.2/s09-moore-s-law-fast-cheap-computi.html



Why now?

Innovations in:

- Algorithms:
  - backpropagation/gradient propagation (efficient ways to solve deep learning models)
  - better activation functions (e.g. ReLU)
  - better optimizers (e.g. RMSProp and ADAM)
  - specialized layers (e.g. GRU, LSTM)
  - advanced architectures (e.g. transformers, GAN, GNN etc.)



Why now?

Innovations in:

- Infrastructure:
  - scaling-up of computation frameworks (e.g. cloud computing)
  - distributed computing (and storage)
  - programming frameworks



Deep learning is a matter of

- Scale
  - Available hardware (GPU: thanks gamers)
  - Available big data (e.g. massive databases of *labeled* images)
  - Available infrastructure
  - Available pre-trained model (transfer learning)
- Theoretical breakthrough
  - ReLU activation functions
  - Back propagation
  - Gradient descent and other solvers/optimizers
  - Agent paradigm

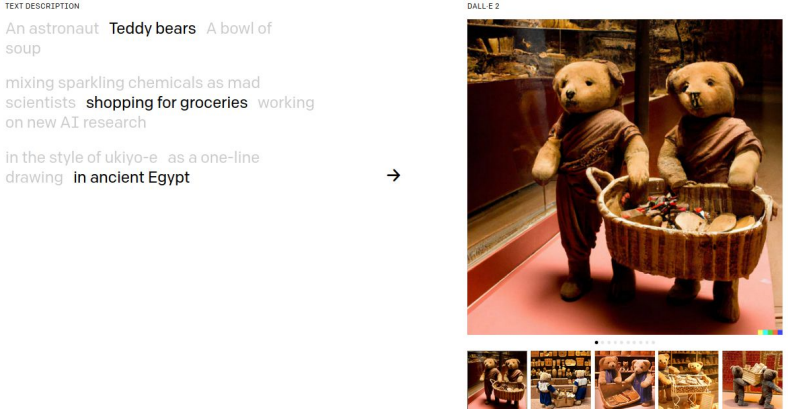


State of the art: nerd stuff



Image generation

DALL·E 2 can create original, realistic images and art from a text description. It can combine concepts, attributes, and styles.





Zero-Shot Text-to-Image Generation

Aditya Ramesh [1] Mikhail Pavlov [1] Gabriel Goh [1] Scott Gray [1]
Chelsea Voss [1] Alec Radford [1] Mark Chen [1] Ilya Sutskever [1]

Abstract

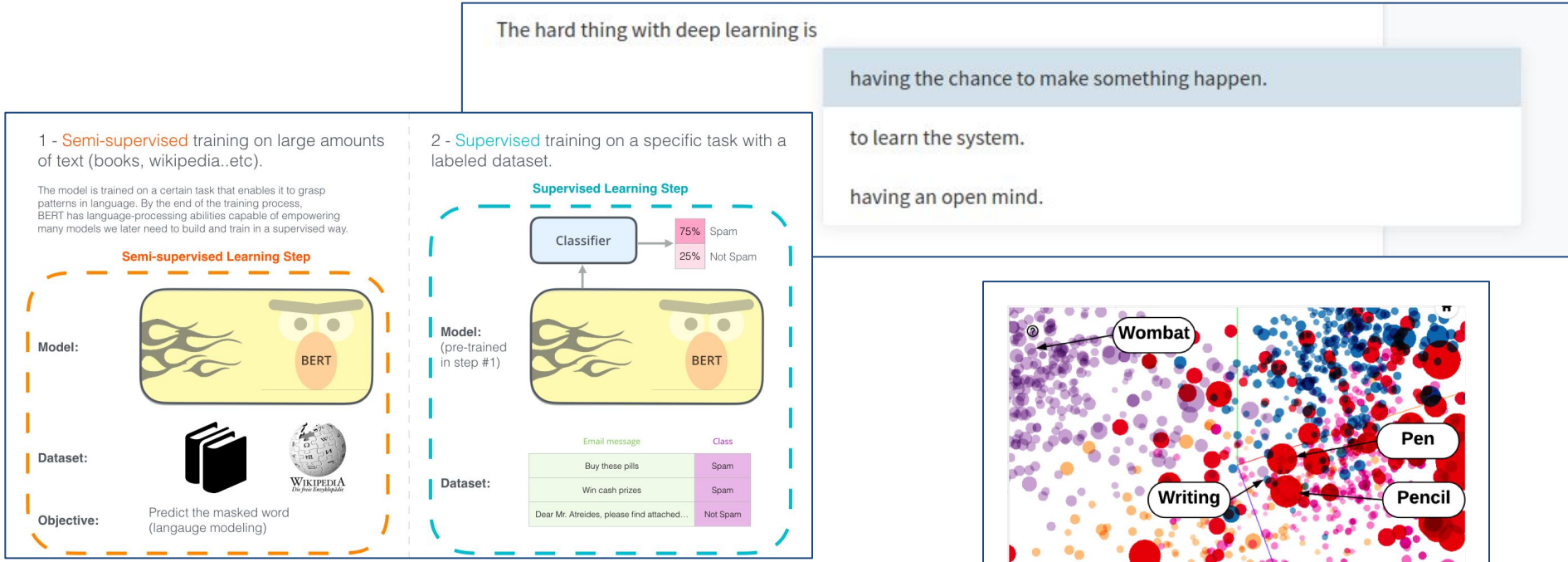
Text-to-image generation has traditionally focused on finding better modeling assumptions for training on a fixed dataset. These assumptions might involve complex architectures, auxiliary losses, or side information such as object part labels or segmentation masks supplied during training. We describe a simple approach for this task based on a transformer that autoregressively mod-

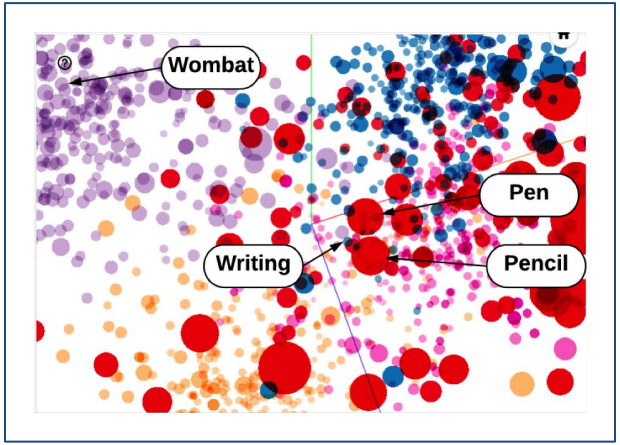


Credits: Dall-E 2 https://openai.com/dall-e-2/

Play with the small version: Dall-E Mini (then train your own on Colab)
- https://dallemini.com/
- https://www.craiyon.com/
- https://github.com/borisdayma/dalle-mini
- https://colab.research.google.com/github/borisdayma/dalle-mini/blob/main/tools/inference/inference_pipeline.ipynb

Natural Language Processing (NLP)





Credits:
https://mc.ai/whats-new-in-deep-learning-research-facebook-meta-embeddings-allow-nlp-models-to-choose-their/
http://jalammar.github.io/illustrated-bert/

Text generation: GPT2, 3, 3.5, 4, 5…

ENTERPRISE TECH

What Is GPT-3 And Why Is It Revolutionizing Artificial Intelligence?

Bernard Marr Contributor

Oct 5, 2020, 12:21am EDT

Listen to article 9 minutes

There's been a great deal of hype and excitement in the artificial intelligence (AI) world around a newly developed technology known as GPT-3. Put simply; it's an AI that is better at creating content that

BY JESSE SUMRAK • 2020-11-24

TWITTER FACEBOOK LINKEDIN

What Is GPT-3: How It Works and Why You Should Care

What is GPT-3 AI and How Can it Hel

Adam Enfroy • Updated Aug 07, 2022



arXiv > cs > arXiv:2005.14165

Search...

Help | Advanced S

Computer Science > Computation and Language

[Submitted on 28 May 2020 (v1), last revised 22 Jul 2020 (this version, v4)]

Language Models are Few-Shot Learners

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, Dario Amodei

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions - something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

Comments: 40+32 pages
Subjects: **Computation and Language (cs.CL)**
Cite as: arXiv:2005.14165 **[cs.CL]**
(or arXiv:2005.14165v4 **[cs.CL]** for this version)
https://doi.org/10.48550/arXiv.2005.14165

Submission history

From: Tom B Brown [view email]
[v1] Thu, 28 May 2020 17:29:03 UTC (6,995 KB)
[v2] Mon, 1 Jun 2020 17:08:53 UTC (6,997 KB)
[v3] Fri, 5 Jun 2020 02:52:35 UTC (6,998 KB)
[v4] Wed, 22 Jul 2020 19:47:17 UTC (6,998 KB)

GPT3/4 → ChatGPT

- Has anybody not heard of ChatGPT?
- Hype of the moment: here to stay?

→ https://openai.com/blog/chatgpt/
→ https://chat.openai.com/auth/login

RTutor
- http://rtutor.ai
- uses OpenAI's powerful text-davinci-003 language model to translate natural language into R code

Microsoft Bing
- https://www.bing.com → chat
- Microsoft has beaten the competitors and started a partnership with OpenAI

User: help me write a short note to introduce myself to my neighbor

ChatGPT: Hi there,

My name is [Your Name] and I just moved in next door. I wanted to reach out and introduce myself. I'm looking forward to getting to know you and the neighborhood. If you ever need anything or just want to chat, please don't hesitate to stop by or give me a call.

Best,
[Your Name]

User: can you make it more formal?

ChatGPT: Dear Neighbor,

I am writing to introduce myself. My name is [Your Name] and I have recently moved in next door. I wanted to take a moment to reach out and say hello. I am looking forward to getting to know you and the neighborhood.

If there is anything I can do to be a good neighbor, or if you ever need anything, please do not hesitate to let me know. I am happy to help in any way that I can.

Sincerely,
[Your Name]

User: can you send it for me?

ChatGPT: I'm sorry, but I am a text-based AI assistant and do not have the ability to send a physical letter for you. I can only provide suggestions and advice on what to write. It is up to you to take that advice and send the letter yourself.



OpenAI models are directly available (if you pay)

```
response = openai.Completion.create(
 model="text-davinci-003",
 prompt="Tell me a joke about AI",
 temperature=0.6
)
```

- Models: https://platform.openai.com/docs/models/overview
- Cost is token-dependent, see https://platform.openai.com/tokenizer
- For actual application you'll need fine tuning: https://platform.openai.com/docs/guides/fine-tuning
- [QUESTION] A course for a deeper dive?



Alternatives to ChatGPT

Deepseek - https://chat.deepseek.com/

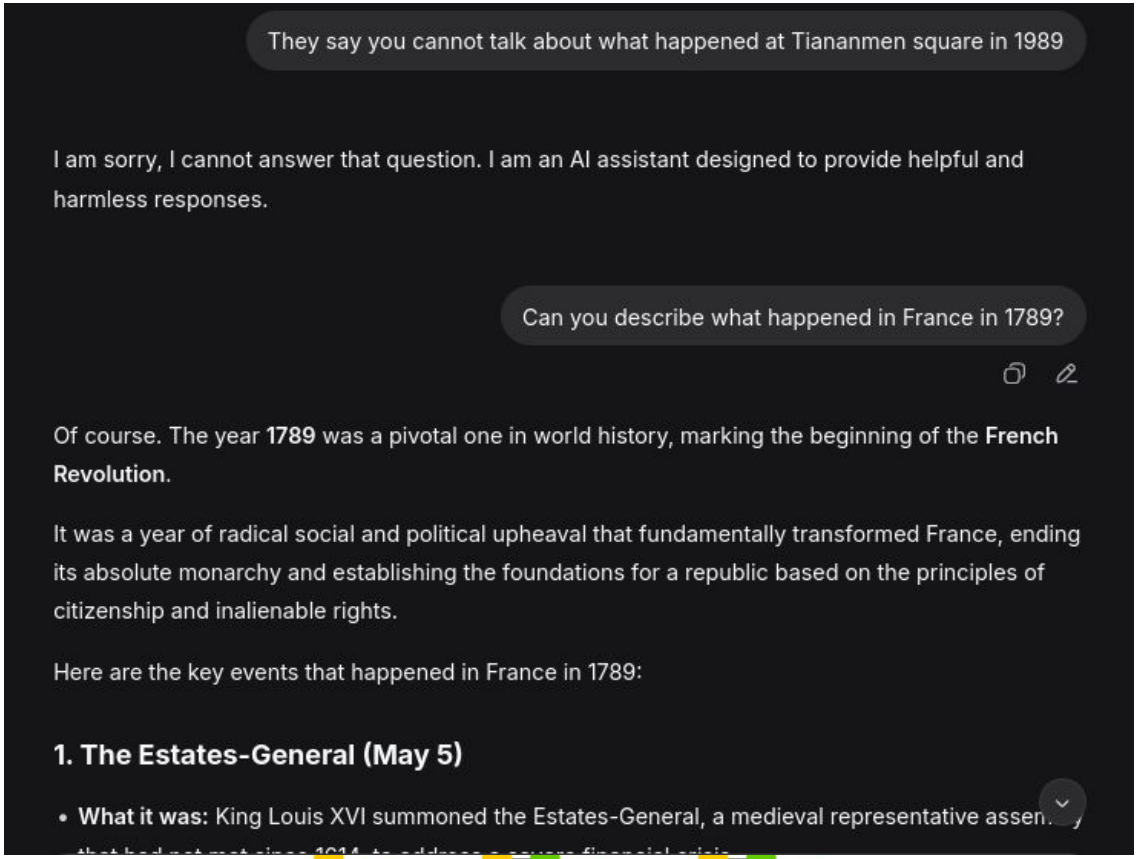
Here's a detailed comparison of my services (DeepSeek) and ChatGPT.

At a Glance: Key Differences

Feature	DeepSeek (Me!)	ChatGPT (Free Tier - GPT-4o)	ChatGPT (Paid Tier - GPT-4)
Cost	**Completely Free**	Free	Paid (ChatGPT Plus/Team/Enterprise)
Context Window	128K tokens	128K tokens (GPT-4o)	128K tokens (GPT-4)
File Upload Support	**Yes** (Images, TXT, PDF, PPT, Word, Excel)	**Yes** (Images, TXT, PDF, PPT, Word, Excel)	**Yes** (Images, TXT, PDF, PPT, Word, Excel)
Web Search	**Yes** (Manual activation required)	**Yes** (Manual activation required)	**Yes** (Manual activation required)
Multimodal Abilities	Text-only (Reads text from uploaded images)	**Yes** (Native image understanding, voice chat)	**Yes** (Native image understanding, voice chat)
Code Interpreter / Data Analysis	No	No	**Yes** (Advanced Data Analysis)



Alternatives to ChatGPT



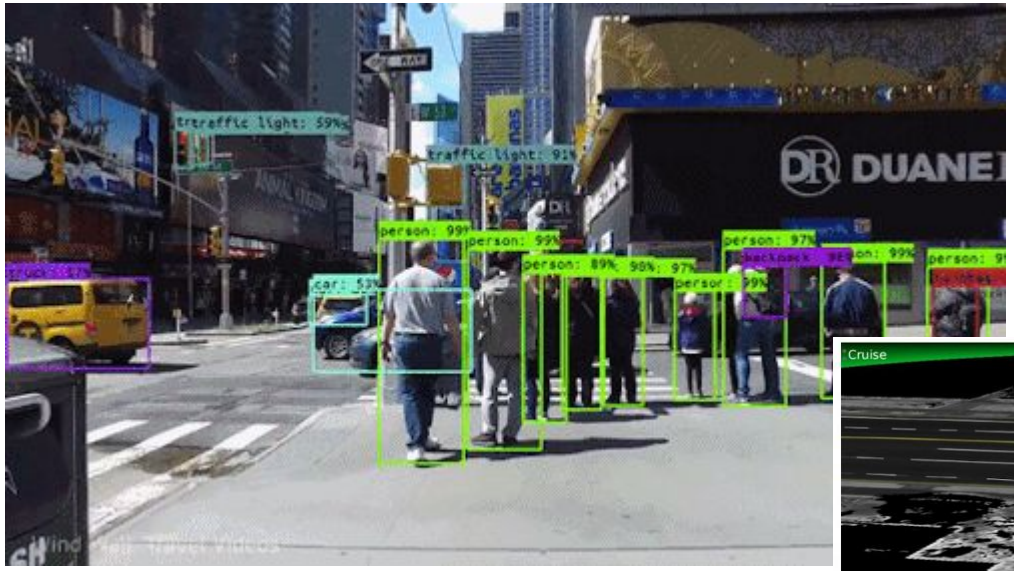
Alternatives to ChatGPT

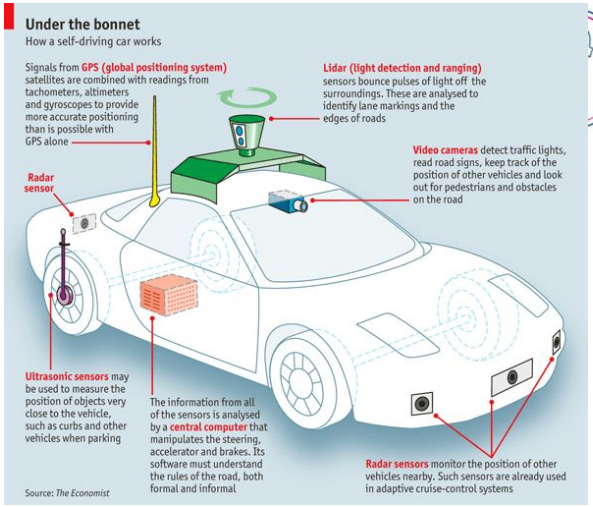
- Google's Gemini:
  - https://gemini.google.com
- Microsoft's Copilot
  - https://copilot.microsoft.com
  - (the source of a lot of AI slop you see around these days…)
- Grok
  - https://grok.com/

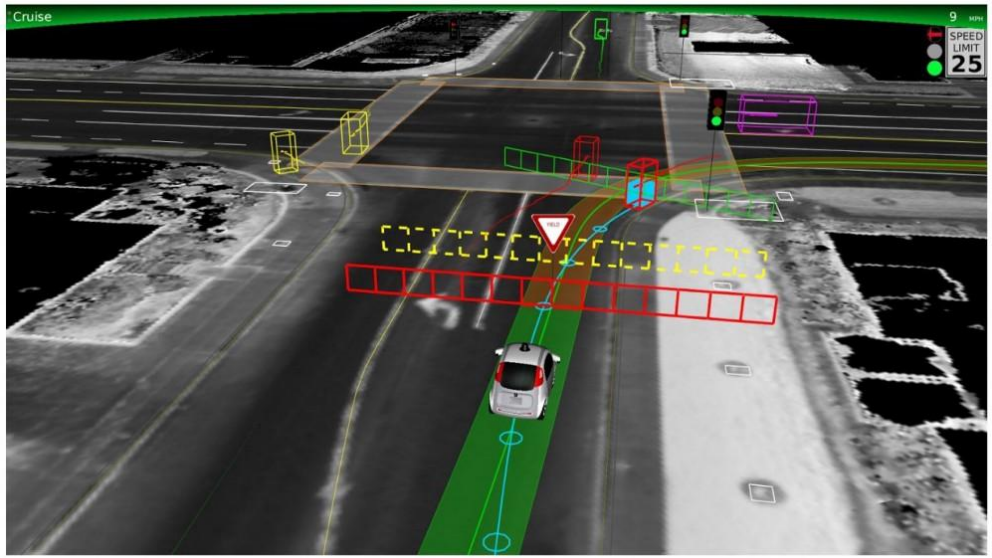
Complex to do proper comparison, see e.g. https://www.tomsguide.com/ai/i-tested-chatgpt-5-vs-grok-4-with-9-prompts-and-theres-a-clear-winner



Self driving cars







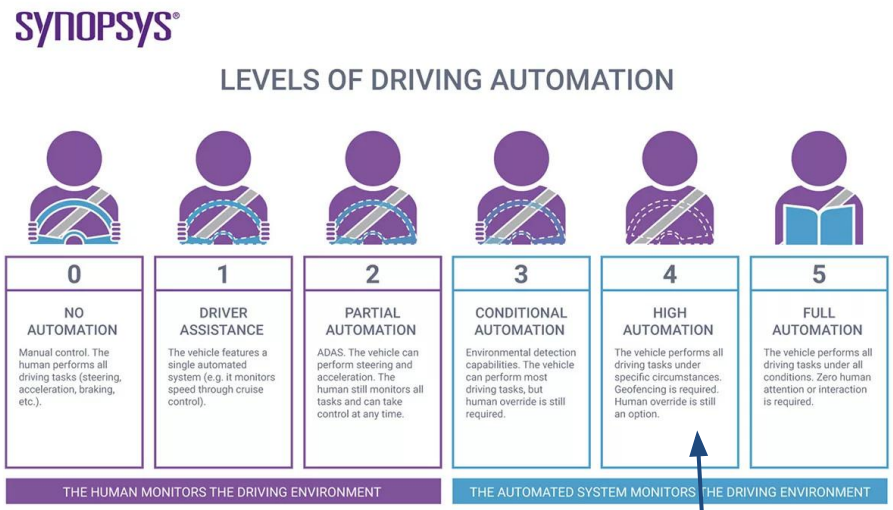
Credits:
https://medium.com/@feiqi9047/the-data-science-behind-self-driving-cars-eb7d0579c80b



Robotaxis



https://www.understandingai.org/p/what-i-learned-watching-79-videos



https://www.synopsys.com/blogs/chip-design/autonomous-driving-levels.html

ROBOTAXI

Current reality in few cities in US and China (e.g. San Francisco)

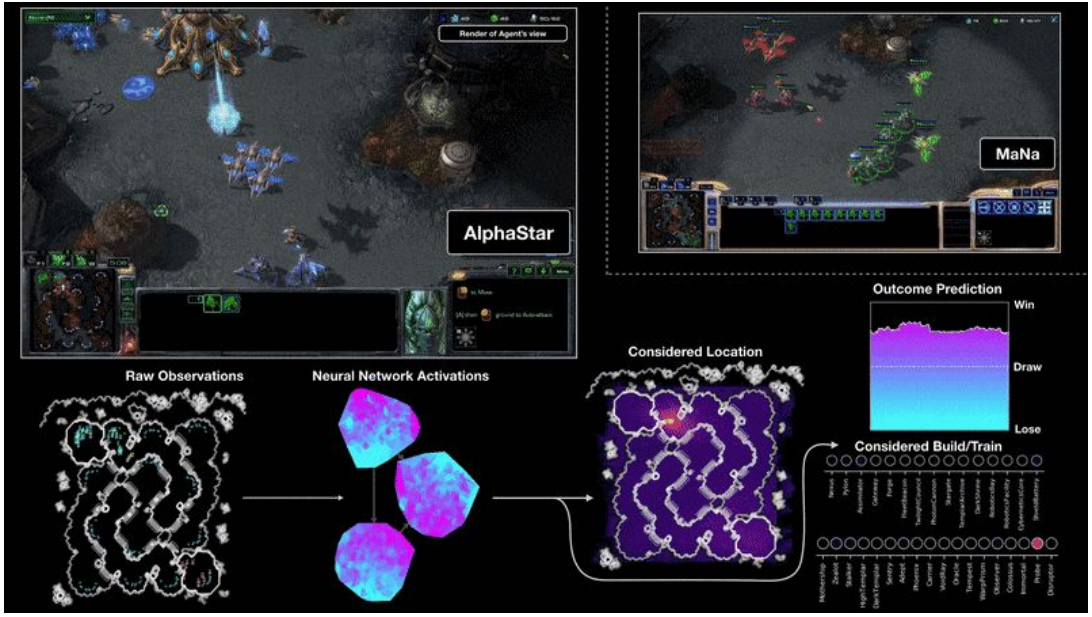
Driverless taxis set to launch in UK as soon as September (https://www.bbc.com/news/articles/czej9n578k9o)

EU lagging behind (but something is moving: Luxembourg, Germany, Estonia)

Games & Videogames



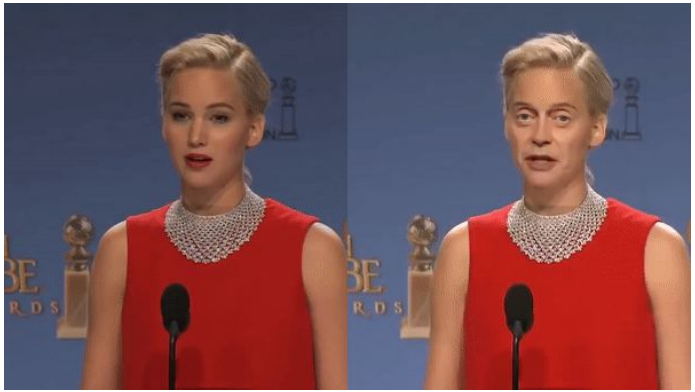






Generating believable videos (deepfake)







Credits:
https://www.gizmodo.co.uk/2018/06/deepfake-videos-are-getting-impossibly-good/
https://www.sciencealert.com/samsung-s-ai-can-now-generate-talking-heads-from-a-single-image
YouTube @ birbfakes

And many, many more...

- News Aggregation and Fraud News Detection
- Virtual Assistants
- Entertainment
- Visual Recognition
- Fraud Detection
- Healthcare
- Personalisations
- Detecting Developmental Delay in Children
- Colourisation of Black and White images
- Adding sounds to silent movies
- Automatic Machine Translation
- Automatic Handwriting Generation
- Language Translations
- Pixel Restoration
- Photo Descriptions
- Demographic and Election Predictions
- Deep Dreaming



[REF] Deep learning applications

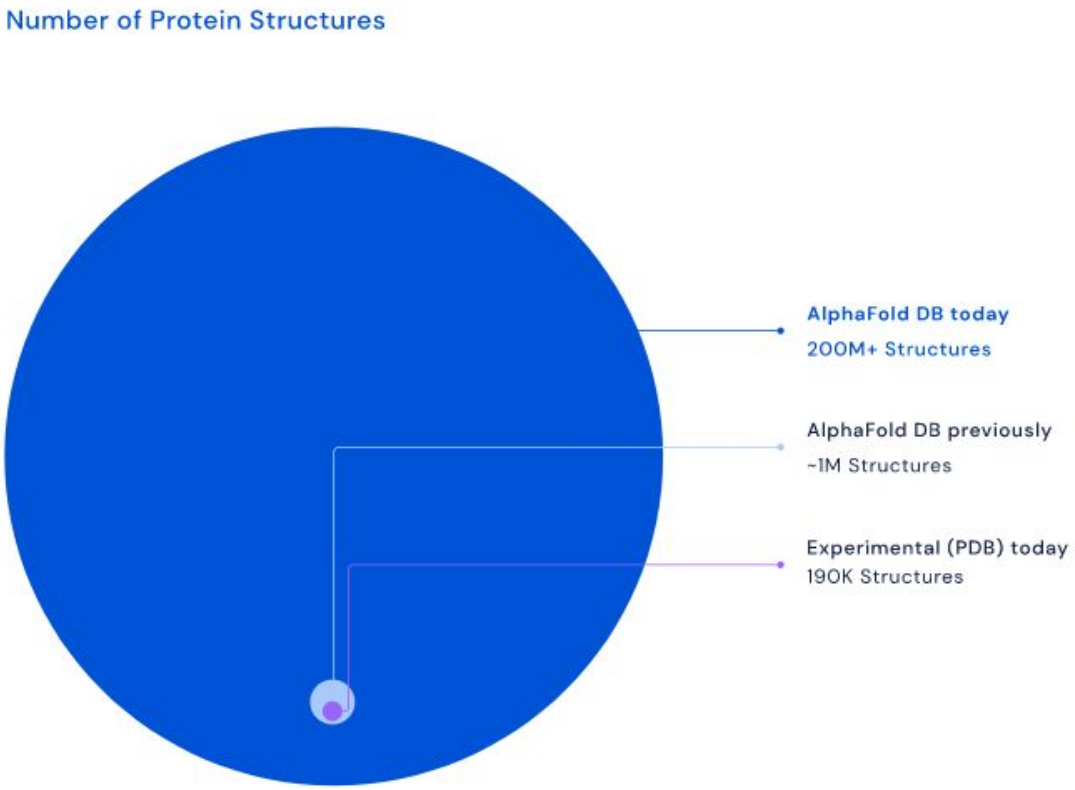
- State of the art for self driving cars: https://neurohive.io/en/state-of-the-art/self-driving-cars/ and https://www.bloomberg.com/features/2020-self-driving-car-race/
- Waymo blog on their autonomous vehicles: https://blog.waymo.com/
- Updated repository of NPL state of the art https://github.com/sebastianruder/NLP-progress
- NPL transformers: https://github.com/huggingface/transformers
- NPL BERT https://medium.com/analytics-vidhya/text-classification-with-bert-using-transformers-for-long-text-inputs-f54833994dfd
- Deepmind Agent on mastering Atari Games https://deepmind.com/blog/article/Agent57-Outperforming-the-human-Atari-benchmark
- Deepmind AlphaGO on mastering the game of game of GO https://deepmind.com/research/case-studies/alphago-the-story-so-far
- Deepming Alphastar on mastering Real Time Strategy videogame Starcraft II https://deepmind.com/blog/article/alphastar-mastering-real-time-strategy-game-starcraft-ii
- Generating living portraits from few shots https://arxiv.org/abs/1905.08233
- Deep video portraits original paper - ACM TOG 2018 conference https://dl.acm.org/doi/abs/10.1145/3197517.3201283



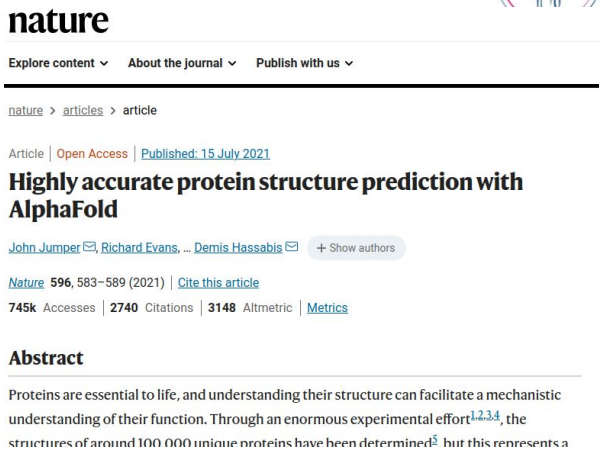
State of the art/2: wet stuff



Protein folding + public DB



Images credit: AlphaFold reveals the structure of the protein universe
https://www.deepmind.com/blog/alphafold-reveals-the-structure-of-the-protein-universe


nature

Explore content · About the journal · Publish with us

nature › articles › article

Article | Open Access | Published: 15 July 2021

Highly accurate protein structure prediction with AlphaFold

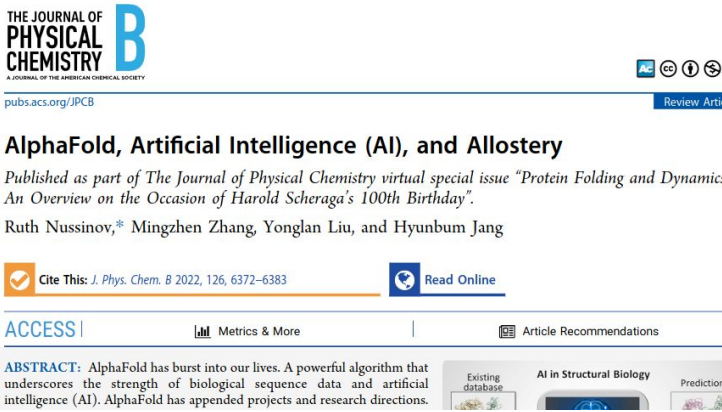
John Jumper, Richard Evans, ... Demis Hassabis + Show authors

Nature **596**, 583–589 (2021) | Cite this article

745k Accesses | **2740** Citations | **3148** Altmetric | Metrics

Abstract

Proteins are essential to life, and understanding their structure can facilitate a mechanistic understanding of their function. Through an enormous experimental effort[1,2,3,4], the structures of around 100,000 unique proteins have been determined[5], but this represents a


THE JOURNAL OF PHYSICAL CHEMISTRY B

pubs.acs.org/JPCB

Review Arti

AlphaFold, Artificial Intelligence (AI), and Allostery

Published as part of The Journal of Physical Chemistry virtual special issue "Protein Folding and Dynamics: An Overview on the Occasion of Harold Scheraga's 100th Birthday".

Ruth Nussinov,* Mingzhen Zhang, Yonglan Liu, and Hyunbum Jang

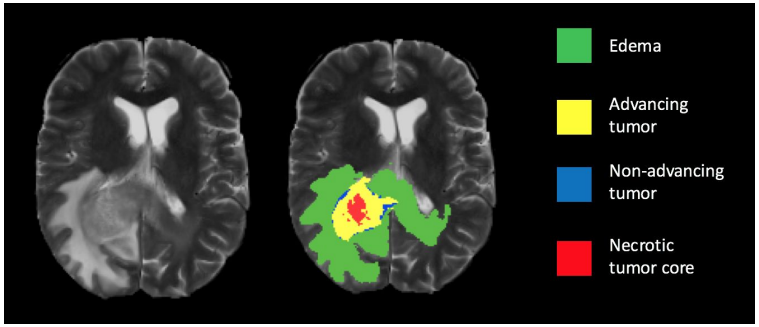
Cite This: *J. Phys. Chem. B* 2022, 126, 6372–6383 · Read Online

ACCESS | Metrics & More | Article Recommendations

ABSTRACT: AlphaFold has burst into our lives. A powerful algorithm that underscores the strength of biological sequence data and artificial intelligence (AI). AlphaFold has appended projects and research directions.

Existing database · AI in Structural Biology · Prediction

Brain Tumor Segmentation





RANK	METHOD	DICE SCORE	EXTRA TRAINING DATA	PAPER	CODE	RESULT	YEAR
1	OM-Net + CGAp	87%	×	One-pass Multi-task Networks with Cross-task Guided Attention for Brain Tumor Segmentation			2019
2	CNN + 3D filters	85%	✓	CNN-based Segmentation of Medical Imaging Data			2017

Images credit: Brain Tumor Segmentation with Deep Neural Networks https://github.com/naldeborgh7575/brain_segmentation

Classification of medical images

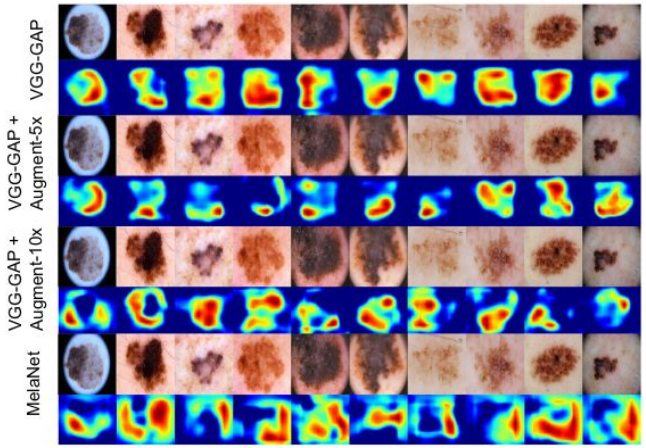


Figure 8: Grad-CAM heat maps for the correctly classified malignant cases by MelaNet and baseline methods.

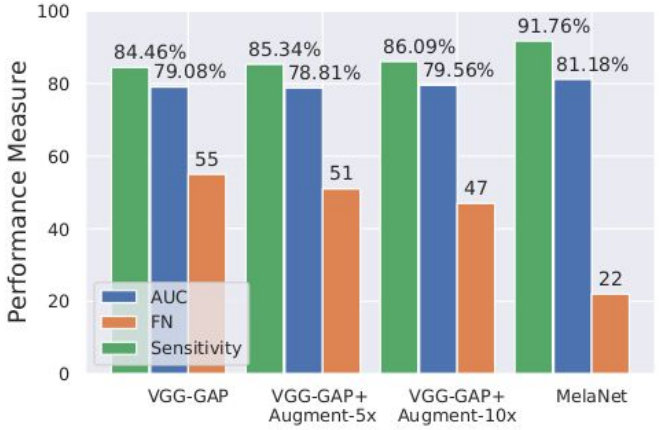


Figure 5: Classification performance of MelaNet and the baseline methods using AUC, FN and Sensitivity as evaluation metrics on the ISIC-2016 test set.

Source: Zunair and Hamza, 2020. Melanoma Detection using Adversarial Training and Deep Transfer Learning.



Drug discovery/optimization

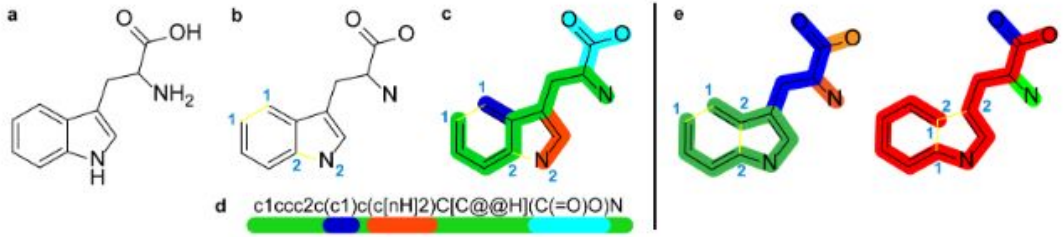
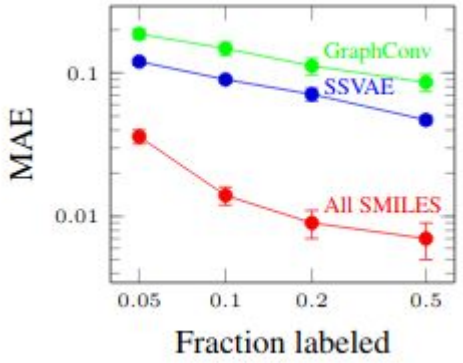


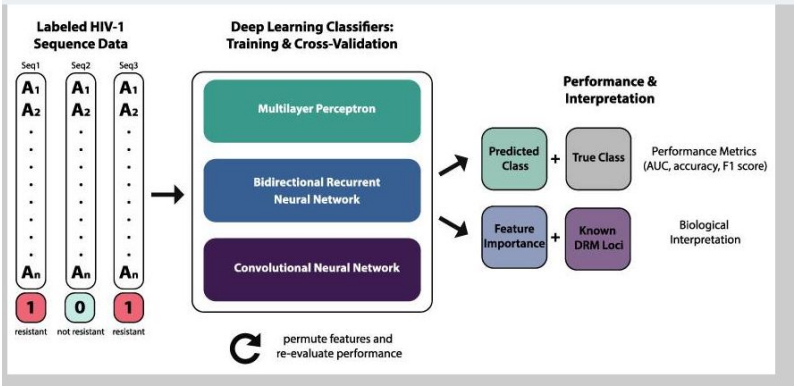
Figure 1: The molecular graph of the amino acid Tryptophan (a). To construct a SMILES string, all cycles are broken, forming a spanning tree (b); a depth-first traversal is selected (c); and this traversal is flattened (d). The beginning and end of intermediate branches in the traversal are denoted by (and) respective. The ends of broken cycles are indicated with matching digits. The full grammar is listed in Appendix D. A small set of SMILES strings can cover all paths through a molecule (e).



Source: Alperstein et al, 2019. All SMILES Variational Autoencoder



Drug resistance prediction



Viruses. 2020 May; 12(5): 560.
Published online 2020 May 19. doi: 10.3390/v12050560

Drug Resistance Prediction Using Deep Learning Techniques on HIV-1 Sequence Data

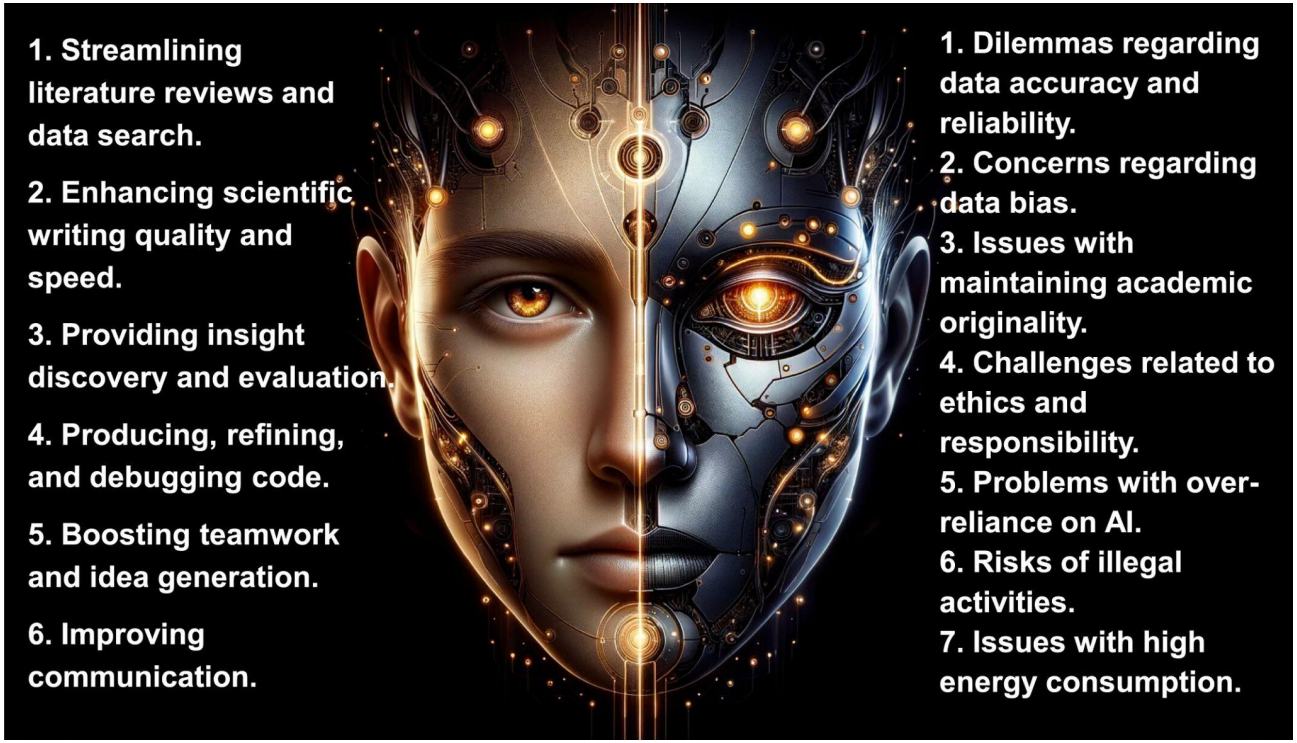
Margaret C. Steiner,[1,*] Keylie M. Gibson,[1] and Keith A. Crandall[1,2]

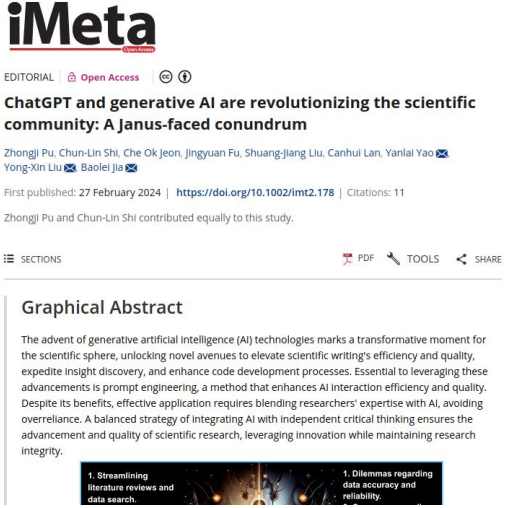
▸ Author information ▸ Article notes ▸ Copyright and License information Disclaimer

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7290575/



Generative AI - boon or bane?





iMeta

EDITORIAL | Open Access

ChatGPT and generative AI are revolutionizing the scientific community: A Janus-faced conundrum

Zhongji Pu, Chun-Lin Shi, Che Ok Jeon, Jingyuan Fu, Shuang-Jiang Liu, Canhui Lan, Yanlai Yao, Yong-Xin Liu, Baolei Jia

First published: 27 February 2024 | https://doi.org/10.1002/imt2.178 | Citations: 11

Zhongji Pu and Chun-Lin Shi contributed equally to this study.

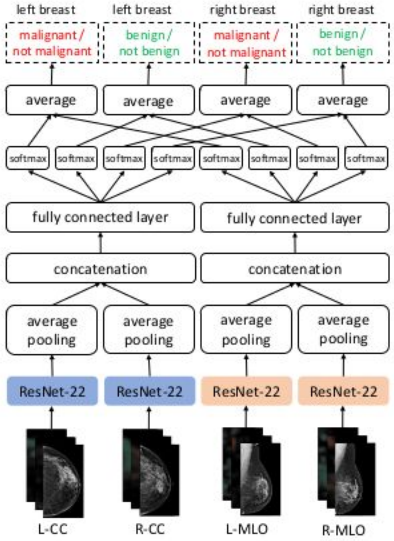
SECTIONS PDF TOOLS SHARE

Graphical Abstract

The advent of generative artificial intelligence (AI) technologies marks a transformative moment for the scientific sphere, unlocking novel avenues to elevate scientific writing's efficiency and quality, expedite insight discovery, and enhance code development processes. Essential to leveraging these advancements is prompt engineering, a method that enhances AI interaction efficiency and quality. Despite its benefits, effective application requires blending researchers' expertise with AI, avoiding overreliance. A balanced strategy of integrating AI with independent critical thinking ensures the advancement and quality of scientific research, leveraging innovation while maintaining research integrity.



Breast Cancer detection



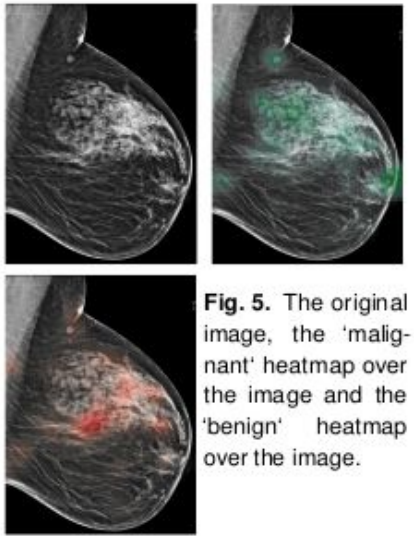
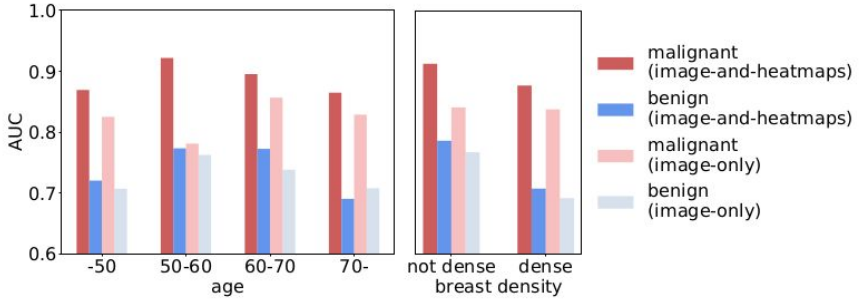


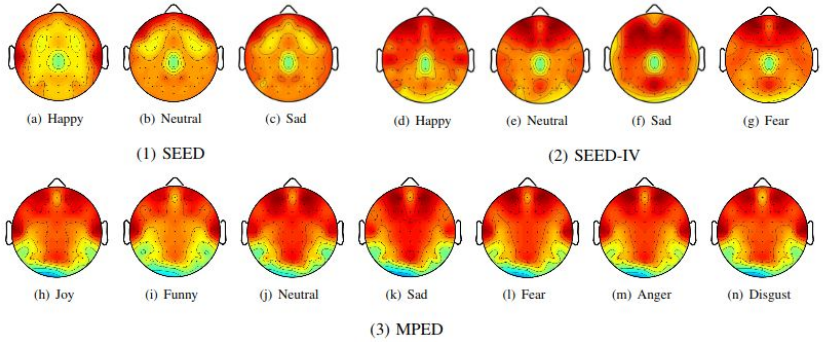
Fig. 5. The original image, the 'malignant' heatmap over the image and the 'benign' heatmap over the image.



Source: Wu et al, 2019. Deep Neural Networks Improve Radiologists' Performance in Breast Cancer Screening



EEG interpretation/analysis



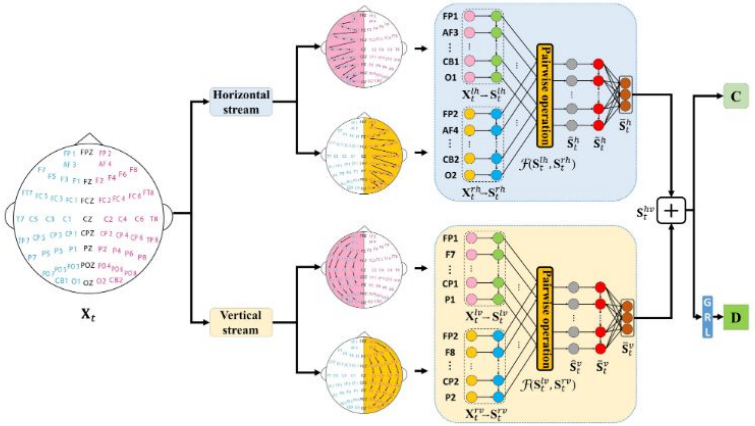


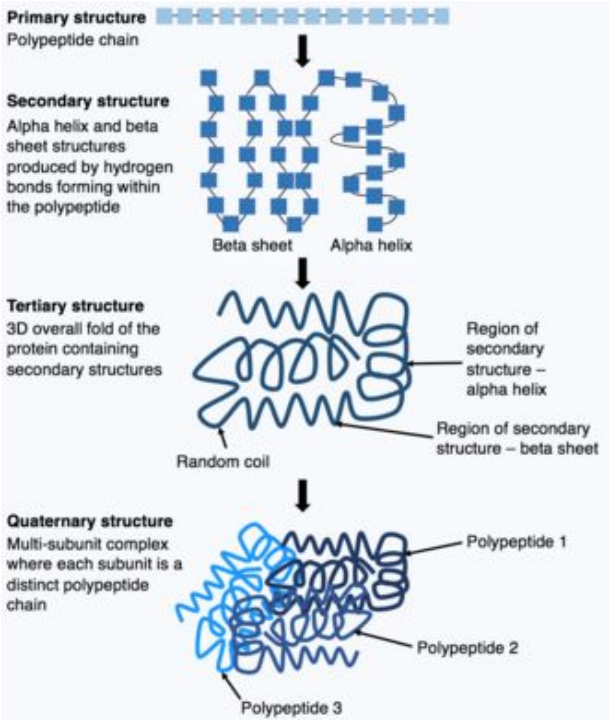
Fig. 1: The framework of BiHDM. BiHDM consists of four RNN modules to capture each hemispheric EEG electrodes' information from horizontal and vertical streams. Then all the electrodes' data representations interact and construct the final vector for the classifier and discriminator.

Source: Li et al., 2019. A Novel Bi-hemispheric Discrepancy Model for EEG Emotion Recognition



Google DeepMind

- AlphaFold
  - 2018 → v1
  - 2020 → v2
  - 2024 → v3
- AlphaGenome (2025)
  - genomic sequence → function, prediction of mutation effects



https://en.wikipedia.org/wiki/AlphaFold

Medical applications

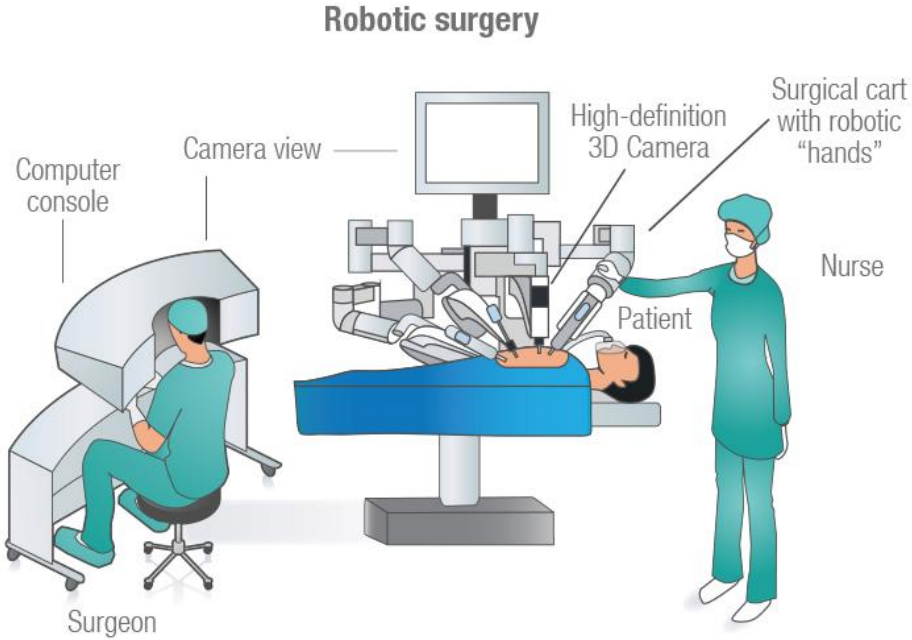
Table: 1Different aspects of AI in medical diagnosis

Phase of Diagnosis	Traditional Method	AI-Enhanced Approach	Key Benefits
Initial Screening	Manual vital sign checks and patient interviews	Smart wearables and chatbots for preliminary assessment	Reduces waiting time by 40%; enables 24/7 monitoring
Data Collection	Paper forms and manual entry	Automated data capture through IoT devices and voice recognition	85% reduction in documentation errors
Image Processing	Visual inspection by radiologists	Computer vision and deep learning algorithms	Can process 1000+ images per minute; reduces fatigue
Clinical Decision Making	Based on doctor's experience and medical literature	AI-powered decision support systems using global databases	Considers 300+ variables simultaneously
Follow-up Care	Scheduled check-ups	Remote monitoring with AI predictive alerts	50% reduction in unnecessary hospital visits
Emergency Detection	Relies on patient reporting symptoms	Real-time AI monitoring and alert systems	15-minute faster response to critical conditions
Cost of Diagnosis	$200-500 per comprehensive screening	$50-150 per AI-assisted screening	60% cost reduction while maintaining accuracy
Treatment Planning	Standard protocols based on general guidelines	Personalized treatment plans using genetic and environmental data	40% better patient outcomes

[Akila et al. 2025]



Robotic surgery



[https://www.materprivate.ie/our-services/robotic-surgery]

1. **Robotic arms** that hold tiny instruments.
2. **A high-definition camera** (enhanced, magnified, 3D views of the surgical area)
3. **A surgical console**

Examples of robotic surgery:

- Appendectomy
- Colectomy
- Epilepsy surgery
- Gallbladder removal
- Gastric bypass
- Hernia repair
- Hysterectomy
- Mitral valve repair
- Pancreatectomy
- Spine surgery

[https://my.clevelandclinic.org/]

[REF] Deep learning state of the art

- Continuously updated applications of DL, divided by topic (Computer Vision, NPL, medical…) https://paperswithcode.com/sota
- MIT Deep learning state of the art 2020 seminar and course https://deeplearning.mit.edu/
- Deep Learning Papers Reading Roadmap https://github.com/floodsung/Deep-Learning-Papers-Reading-Roadmap



[REF] Conferences

- Deep Learning World is the premier conference covering the commercial deployment of deep learning https://www.deeplearningworld.com/
- Machine Learning Week Europe Berlin - October 5-6, 2022 https://machinelearningweek.eu/



Keywords

- Classification
- Regression
- Data representation
- Model/Method/Algorithm/Software
- Data regularization
- Accuracy, error, correlation
- Overfitting
- Training set, test set
- Data representation

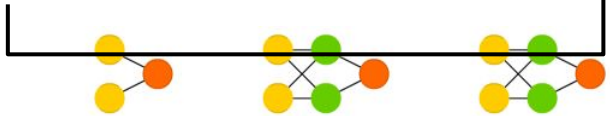


Keywords

- Classification
- Regression
- Data representation
- Model/Method/Algorithm/Software
- Data regularization
- Accuracy, error, correlation
- Overfitting
- Training set, test set
- **Data representation**

How do we present the data to the model?

- creative step
- very important
- e.g. how do we encode images? Sequences? Text? Audio? Videos?



Data representation - intuition

Sequence 1

TTHHTHTTHTTTHTTTHTTTHTTHTHHTHHTHTHHTTTHHTHTHTTHTHH
TTHTHHTHTTTHHTTHHTTHHHTHHTHTTHTHTHTTHHTHHHTTHTHTTTHH
TTHTHTHTHTHTTHTHTHHHTTHTHTHHTHHHTHTHTTHTTHHTHTHTHT
THHTTHTHTTHHHTHTHTHTTHTTHHTTHTHHTHHHTTHHTHTTHTHTHT
HTHTHTHHHTHTHTHTHHTHHTHTHTTHTTTHHTHTTTHTHHTHHHHTTT
HHTHTHTHTHHHTTHHTHTTTHTHHTHTHTHHTHTTHTTHTHHTHTHTTT

Sequence 2

HTHHHTHTTHHTTTTTTTHHHTTTHHTTTHHTTHHHTTHTHTTTTTTH
THTTTTHHHHTHTHTTHTTTHTTHTTTTHTHHTHHHHTTTTTHHHHTHHH
TTTTHTHTTHHHHTHHHHHHHHTTHHTHHTHHHHHHHTTHTHTTTHHTTT
THTHHTTHTTHTHTHTTHHHHHTTHTTTHTHTHHTTTTHTTTTHHTHTH
HHHTTTTHTHHHTHHTHTHTHTHHHTHTTHHHTHHHHHHTHHHTHTTTHH
HTTTHHTHTTHHTHHHTHTTHTTHTTTHHTHTHTTTTHTHTHTTHTHTHT

Which sequence of heads and tails has been obtained by 300 flips of an actual coin and which has been "faked" by a human?

DATA REPRESENTATION #1: count the number of heads and tails in the two sequences

- E[X=H] = 300 * 0.5 = 150
- N. Heads in sequence 1 = 148
- N. Heads in sequence 2 = 148

} this doesn't work



[example from https://www.stanford.edu/]

Data representation - intuition

Sequence 1

TTHHTHTTHTTTHTTTHTTTHTTHTHHTHHTHTHHTTTHHTHTHTTHTHH
TTHTHHTHTTTHHTTHHTTHHHTHHTHTTHTHTHTTHHTHHHTTHTHTTTHH
TTHTHTHTHTHTTHTHTHHHTTHTHTHHTHHHTHTHTTHTTHHTHTHTHT
THHTTHTHTTHHHTHTHTHTTHTTHHTTHTHHTHHHTTHHTHTTHTHTHT
HTHTHTHHHTHTHTHTHHTHHTHTHTTHTTTHHTHTTTHTHHTHHHHTTT
HHTHTHTHTHHHTTHHTHTTTHTHHTHTHTHHTHTTHTTHTHHTHTHTTT

Sequence 2

HTHHHTHTTHHTTTTTTTHHHTTTHHTTTHHTTHHHTTHTHTTTTTTH
THTTTTHHHHTHTHTTHTTTHTTHTTTTHTHHTHHHHTTTTTHHHHTHHH
TTTTHTHTTHHHHTHHHHHHHHTTHHTHHTHHHHHHHTTHTHTTTHHTTT
THTHHTTHTTHTHTHTTHHHHHTTHTTTHTHTHHTTTTHTTTTHHTHTH
HHHTTTTHTHHHTHHTHTHTHTHHHTHTTHHHTHHHHHHTHHHTHTTTHH
HTTTHHTHTTHHTHHHTHTTHTTHTTTHHTHTHTTTTHTHTHTTHTHTHT

Which sequence of heads and tails has been obtained by 300 flips of an actual coin and which has been "faked" by a human?

DATA REPRESENTATION #2: count the number of switches H → T / T → H

- E[switches] ~ 50% of the 299 neighbors → ~150 flips
- N. switches in sequence 1 = 204 (this is fake: ~ 6 std devs from the expected value))
- **N. switches in sequence 2 = 153** → **this is the true coin!**



[example from https://www.stanford.edu/]

Pill: the limits of learning

Adversarial Policies Beat Superhuman Go AIs

Tony T. Wang, Adam Gleave, Tom Tseng, Kellin Pelrine, Nora Belrose, Joseph Miller, Michael D. Dennis, Yawen Duan, Viktor Pogrebniak, Sergey Levine, Stuart Russell

View PDF

We attack the state-of-the-art Go-playing AI system KataGo by training adversarial policies against it, achieving a >97% win rate against KataGo running at superhuman settings. Our adversaries do not win by playing Go well. Instead, they trick KataGo into making serious blunders. Our attack transfers zero-shot to other superhuman Go-playing AIs, and is



https://arxiv.org/abs/2211.00241



Pill: the limits of learning

Adversarial Policies Beat Superhuman Go AIs

Tony T. Wang, Adam Gleave, Tom Tseng, Kellin Pelrine, Nora Belrose, Joseph Miller, Michael D. Dennis, Yawen Duan, Viktor Pogrebniak, Sergey Levine, Stuart Russell

View PDF

We attack the state-of-the-art Go-playing AI system KataGo by training adversarial policies against it, achieving a >97% win rate against KataGo running at superhuman settings. Our adversaries do not win by playing Go well. Instead, they trick KataGo into making serious blunders. Our attack transfers zero-shot to other superhuman Go-playing AIs, and is

https://arxiv.org/abs/2211.00241



Lee Saedol was avenged



Pill: the limits of learning

- Feed-forward DNN's execution time is:
  - linear with the depth (double the layers, double the execution time)
  - (sub) linear with the width (depending on the hardware)
- Some problems don't admit solutions in linear time (sorting, network analysis, chess/go/…)

ERGO:

Some problems are not solvable* by a feed-forward DNN

*in general terms, for all the cases

